

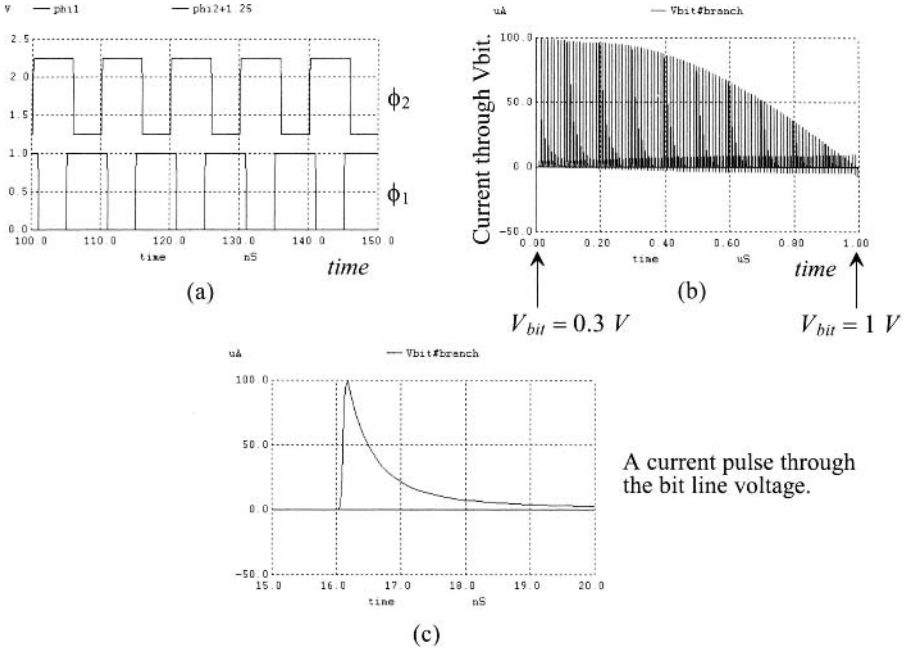


Figure 17.11 The operation of the circuit in Fig. 17.10.

V_{bit} is 1 V (= V_{DD}). Let's use Eq. (17.11) to calculate the charge supplied by C_{cup} when V_{bit} is 300 mV

$$Q_{cup} = 100 \text{ fF} \cdot (1 - 0.3) = 70 \text{ fC} \quad (17.14)$$

Figure 17.11c shows the pulse of current that is dumped into V_{bit} at the beginning of the simulation. To estimate the amount of charge in this pulse, we take the amplitude and multiply it by an estimate of the pulse's width. That is, $100 \mu\text{A} \cdot 0.7 \text{ ns} = 70 \text{ fC}$ or the same result as seen in Eq. (17.14).

To validate Eq. (17.12), we can add M4 from Fig. 17.9 to the circuit seen in Fig. 17.10. The only parameter we need to calculate before simulating is the value of V_{REF} . Using Eq. (17.13) and knowing $V_{THP} = 280 \text{ mV}$, we see that if we set V_{REF} to a large voltage, Q_{cup} gets small. If, for example, $V_{REF} = V_{DD} - V_{THP}$ (= 750 mV in this example), Q_{cup} goes to zero. We can write

$$V_{REF} < V_{DD} - V_{THP} \quad (17.15)$$

Let's use a V_{REF} of $V_{DD}/2$ or 500 mV. Figure 17.12 shows the simulation results. Note, in Fig. 17.9, that we place M4 in its own well. Since we care about how the threshold voltage varies, we've eliminated the body effect in this device. The simulation results are seen in Fig. 17.12. In this simulation we've swept the bit line voltage from 0.3 to 0.6 V. Comparing the plot to Fig. 17.11b, we see much better linearity (the charge added to the bit line doesn't change with bit line voltage). The cost for this improvement is more limited voltage swing on the bit line. ■

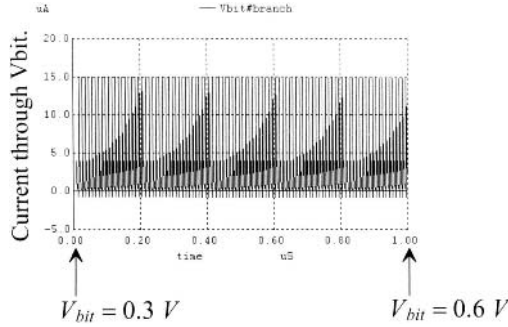


Figure 17.12 How the charge supplied to V_{bit} becomes linear if we add M4, from Fig. 17.9, to the simulation in Fig. 17.10. Note the reduced bit line voltage swing and the reduction in the current pulse amplitudes.

Incomplete Settling

If we zoom in on the current pulses in Fig. 17.12, we get a view like the one seen in Fig. 17.13. The glitches at 30 and 40 ns are the result of the ϕ_2 clock going high and M2 shutting off. Notice how the current through M4 (and V_{bi}) isn't zero when M2 shuts off. Using our water analogy, this is equivalent to stopping the pouring of the water out of the cup before the cup is empty. This incomplete emptying of the cup or capacitor is termed *incomplete settling*. The currents in the circuit haven't gone to zero before the clock transitions. The parasitic capacitances on the sources of M3/M4 continue to discharge through V_{bit} after M2 shuts off, resulting in nonzero current. Incomplete settling has the effect of making the capacitor, C_{cup} , appear smaller. It won't affect the linearity of the sense but it can affect the maximum value of the sensed current. To eliminate the incomplete settling behavior we can increase the width of M4. This results in the source of M4 being held closer to $V_{REF} + V_{THP}$ when M2 and M3 turns on. We could also slow the clock frequency down to ensure that the circuit settles. Because the amount of charge transferred from C_{cup} is constant from one clock cycle to the next, the effects of incomplete settling on DSM are simply a limit on the maximum current that can be removed from the bit line and a modification to the digital output for the absolute value of the input signal, that is, Eq. (17.2) (a smaller value of fed-back signal size is used).

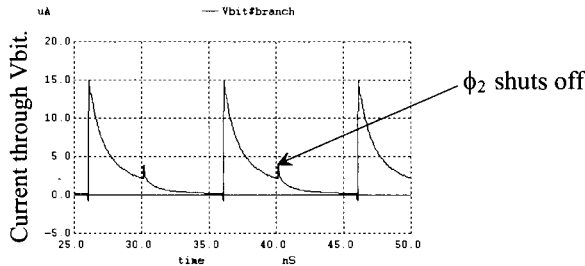


Figure 17.13 The incomplete settling in Fig. 17.12.

17.2 Sensing Resistive Memory

Let's apply the techniques we've just discussed to sensing a resistive memory cell. Figure 17.14a shows a schematic of one-transistor, one-resistor (1T1R) memory cell. Notice the similarity to the 1T1C DRAM memory cell seen in Fig. 16.9. *Ideally* the resistor is either zero ohms (which we'll call *programmed*) or infinite (which we'll call *erased*), Fig. 17.14b. To sense in this ideal case, we simply precharge the bit line to V_{DD} . When the word line is driven high, the access device turns on, resulting in the bit line either moving to $V_{DD}/2$ (if the cell is programmed) or not moving at all (if the cell is erased). In a practical circuit, however, the cell's resistance will be nonzero and not infinite (say $10\text{k}\Omega$ to $100\text{k}\Omega$ as seen in Fig. 17.14c).

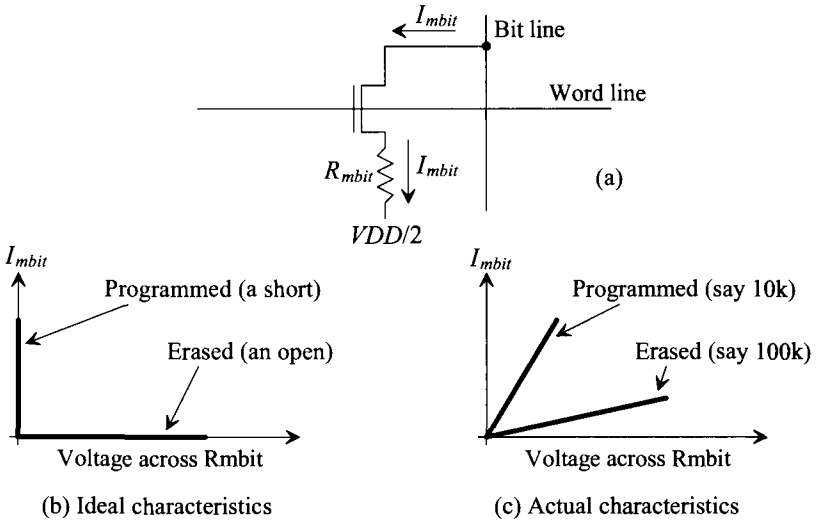


Figure 17.14 The one-transistor, one-resistor (1T1R) RAM memory cell.

The Bit Line Voltage

Consider the block diagram of a DSM seen in Fig. 17.15 with a resistive memory cell. To keep the schematic simpler, we won't include the access MOSFET. We know from the previous discussions that the DSM tries to hold the bit line at precisely the reference voltage used by the comparator. In Fig. 17.15 if the reference voltage used by the comparator is $V_{DD}/2$, then the current that flows through the memory resistor is zero. To avoid this, we might use a reference that is offset from $V_{DD}/2$ by V_{os} . The current that flows in the resistor is then

$$I_{mbit} = \frac{V_{os}}{R_{mbit}} \quad (17.16)$$

In the ideal condition, the erased resistor results in zero I_{mbit} . The output of the DSM is then simply a string of zeroes (we never have to add current to the bit line because none is leaving). If the resistor is programmed, the bitline gets pulled down towards $V_{DD}/2$. The current source can't supply enough charge to pull the bit line up and the DSM output is a constant string of ones (yes, add current).

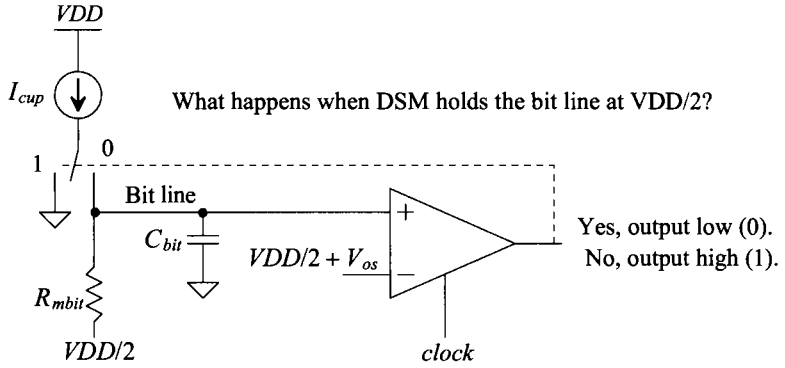


Figure 17.15 Sensing a resistive memory cell using DSM. Notice that we've eliminated the inverter from Fig. 17.8 by switching the input terminals of the comparator.

Adding an Offset to the Comparator

It's important to minimize the number of reference voltages used in a memory sense scheme. We'll use VDD and $VDD/2$ and then design a comparator with a built-in offset voltage. The comparator design is seen in Fig. 17.16 (see Figs. 16.32 and 16.35 from the last chapter). Simulation results showing the offset are seen in Fig. 17.17. When $In+$ gets 50 mV above $In-$, the output of the comparator changes states. It's possible to design the comparator to simplify the sense circuitry, e.g., clock the comparator on the falling edge of ϕ_2 , remove the NAND gates on the output, and combine M2 and M3 in Fig. 17.9. In a

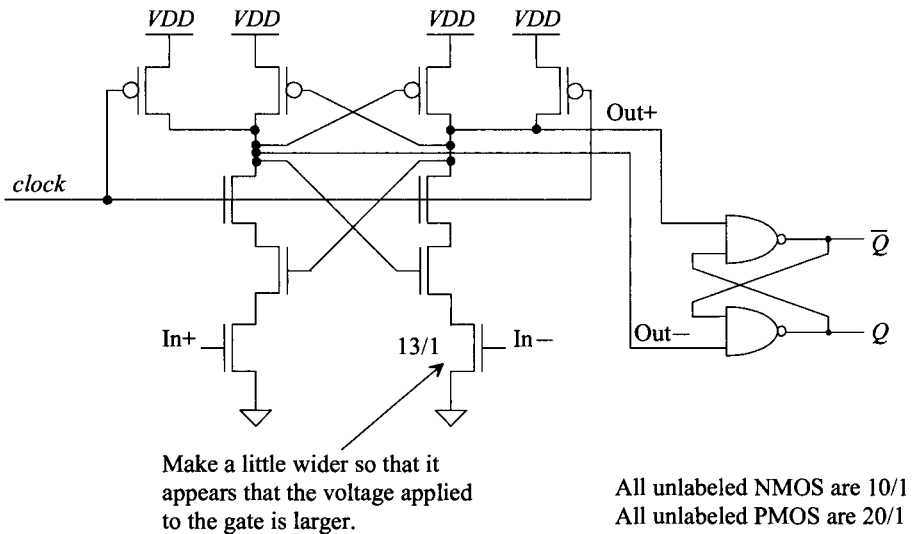


Figure 17.16 Designing a clocked comparator with a built-in offset.

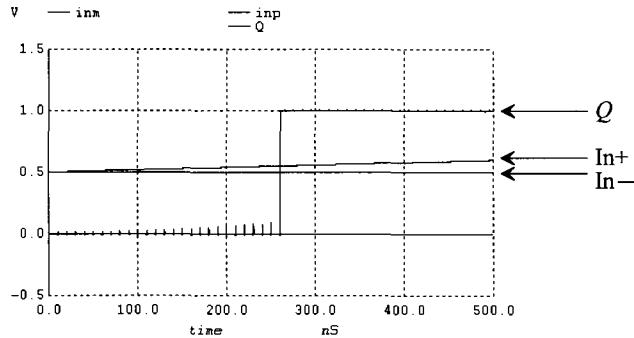


Figure 17.17 The output of the comparator switching states when the + input is 50 mV above the - input.

production part we might want to do this. However, here, where we want to minimize the glitches that may turn some switches inadvertently partially on, we don't try to simplify the circuitry (see problems at the end of the chapter).

Schematic and Design Values

The schematic for a DSM sensing circuit for resistive memories is seen in Fig. 17.18. The capacitance associated with the bit line is shown explicitly (even though it's a parasitic associated with the line). When we are sensing the value of a resistive cell, a word line goes high and connects the resistance, R_{mbit} , to the digit line. The comparator, with its built-in offset, tries to hold the bit line, through the feedback loop, at $VDD/2 + V_{OS}$. The

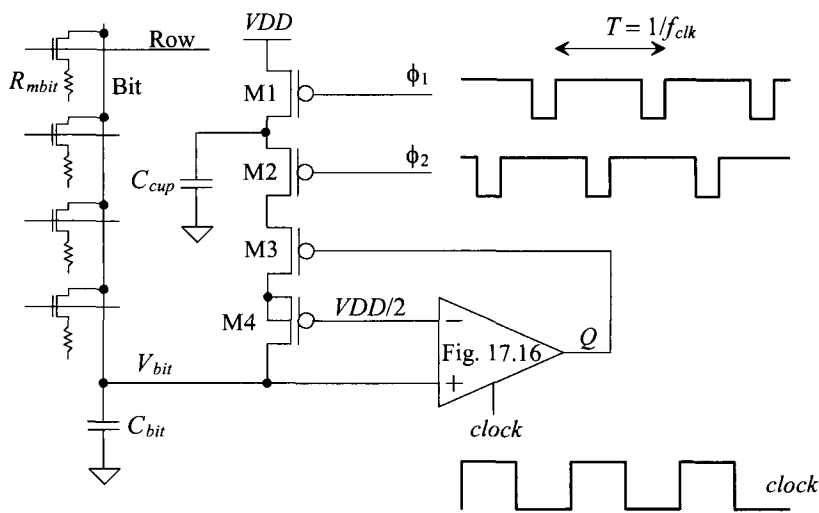


Figure 17.18 Sensing circuit for resistive memory.

current through the cell is then given by Eq. (17.16). Again, if N is the total number of times the DSM is clocked and M is the number of times M3 is enabled so that C_{cup} can dump its charge into the bit line, then we can write (reviewing Eqs. [17.3] to [17.7], [17.12], and [17.16])

$$\frac{V_{OS}}{R_{mbit}} = Q_{cup} \cdot \frac{M}{N} \cdot \frac{1}{T} = (VDD - VDD/2 - V_{THP}) \cdot C_{cup} \cdot \frac{M}{N} \cdot \frac{1}{T} \quad (17.17)$$

or

$$R_{mbit} = \frac{V_{OS} \cdot T}{(VDD/2 - V_{THP}) \cdot C_{cup} \cdot \frac{M}{N}} \quad (17.18)$$

Notice that if R_{mbit} goes to zero (programmed), the output of the DSM is always low (M is always a 1). If R_{mbit} is infinite, then the output of the DSM is always high (M is always a 0). To design the DSM, we need to pick an R_{mbit} value that separates what we define as a low resistance (a programmed state) from what we define as a high resistance (an erased state). Let's use 50k. When the resistance is 50k, then $M = N/2$ (half of the clock cycles the output of the DSM goes high). If $M > N/2$, then the cell is programmed. If $M < N/2$, the cell is erased. If V_{OS} is 50 mV, $VDD = 1$ V, $V_{THP} = 280$ mV, and $f_{clk} = 100$ MHz then

$$C_{cup} = \frac{0.05}{0.22 \cdot 50k \cdot \frac{1}{2}} \cdot 10 \text{ ns} = 90 \text{ fF} \quad (17.19)$$

Since offset won't be precisely 50 mV, we'll round this up to 100 fF (and it will vary with process runs). To estimate the value of the resistor given both N and M , we use Eq. (17.18)

$$R_{mbit} \approx 25k \cdot \frac{N}{M} \quad (17.20)$$

Again, this is an approximation since our value for the offset won't be exactly 50 mV. Note that the minimum value of resistor we can sense occurs when $N = M$ ($R_{mbit} = 25k$).

One more thing that we must calculate before simulating the design is the variation of the bit line voltage (see Eq. [17.8]). The minimum voltage on the bit line is $VDD/2$ (when R_{mbit} is small). The maximum voltage will be $VDD/2 + V_{os} + \Delta V_{bit}$ (where the last term is the maximum change in the bit line voltage). To determine ΔV_{bit} , we can write (knowing charge must be conserved)

$$C_{cup} \cdot (VDD - VDD/2 - V_{THP}) = \Delta V_{bit} \cdot C_{bit} \quad (17.21)$$

or

$$\Delta V_{bit} = \frac{C_{cup}}{C_{bit}} \cdot (VDD/2 - V_{THP}) \quad (17.22)$$

Using the numbers above with a C_{bit} of 500 fF gives a $\Delta V_{bit} = 37 \text{ mV}$. Note that we were very concerned about how the bit line voltage change affected the amount of charge supplied by C_{cup} (see Figs. 17.8, 17.9, and the associated discussion). Here (and Eq. [17.16]) we aren't concerned with how changes in the bit line voltage affect the current through R_{mbit} . The average current through R_{mbit} is V_{OS}/R_{mbit} . The DSM holds the bit line, on average, at $VDD/2 + V_{OS}$. The charge supplied by C_{cup} , however, is not constant on average (as discussed earlier), and that is why M4 was added to the DSM.

Figure 17.19 shows the DSM sensing circuit's output for various values of R_{mbit} . Let's compare the theoretical estimate, that is, Eq. (17.20) to the simulated results. In all of the simulations seen in Fig. 17.19 we clock the DSM 50 times (a 100 MHz clock for 500 ns). In (a), with $R_{mbit} = 25k$, we get an output of 14 zeroes and 36 ones (= M). We can then write

$$R_{mbit} = 25k \cdot \frac{50}{36} = 35k \text{ (actual value 25k)}$$

For (b), we see $M = 17$ so

$$R_{mbit} = 25k \cdot \frac{50}{17} = 73k \text{ (actual value 50k)}$$

and for (c) and (d),

$$R_{mbit} = 25k \cdot \frac{50}{10} = 125k \text{ (actual value 100k)}$$

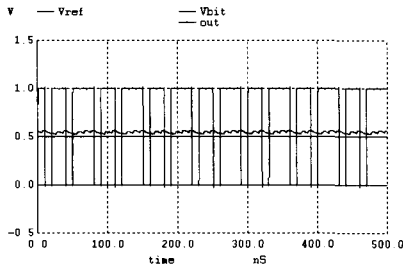
$$R_{mbit} = 25k \cdot \frac{50}{6} = 208k \text{ (actual value 200k)}$$

Notice that as we increase the value of R_{mbit} , we start to get nonlinear. The maximum finite value we can sense with 50 clock pulses is

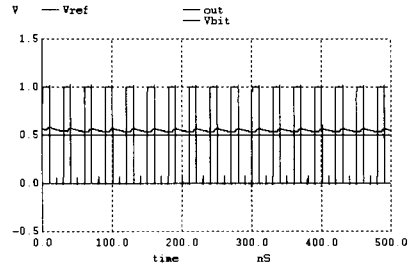
$$R_{mbit,max} = \frac{V_{OS} \cdot T \cdot N}{(V_{DD}/2 - V_{THP}) \cdot C_{cup}} \text{ (= 1.25M here)} \quad (17.23)$$

Again, the minimum resistance is

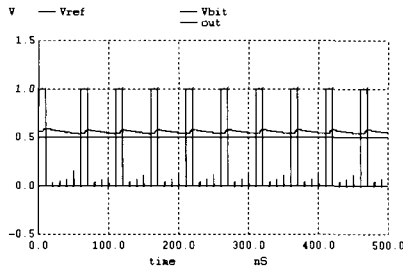
$$R_{mbit,min} = \frac{V_{OS} \cdot T}{(V_{DD}/2 - V_{THP}) \cdot C_{cup}} \text{ (= 25k here)} \quad (17.24)$$



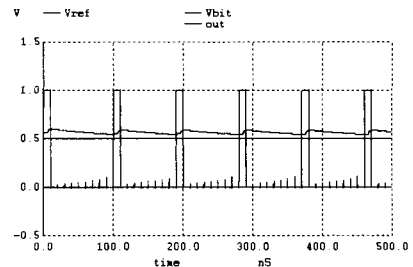
(a) $R_{mbit} = 25 \text{ k}\Omega$



(b) $R_{mbit} = 50 \text{ k}\Omega$



(c) $R_{mbit} = 100 \text{ k}\Omega$



(d) $R_{mbit} = 200 \text{ k}\Omega$

Figure 17.19 Outputs of the DSM for various R_{mbit} values.

Notice that we get good linearity with (actual) resistance values ranging from 25k to 75k. To sense larger values of resistances with good linearity, we must clock the DSM sensing circuit more times (increase N). However, if we are simply trying to separate a “big” resistor from a “small” resistor, we see that the scheme works very well.

A Couple of Comments

We might wonder why the simulated values are different from the actual values. The answers come from incomplete settling (making C_{cup} appear smaller than it actually is) and from the fact that the offset voltage is not precisely 50 mV. For the incomplete settling problem we can clock the circuit slower or try using a larger width for M4. For the offset voltage problem we might try to generate a precise reference voltage ($V_{DD}/2 + V_{OS}$) for the comparator. Practically, this could lead to problems. In a real memory system, the voltages are noisy. Reviewing the designs in Figs. 17.15 and 17.18, we see that if there is noise on $V_{DD}/2$ it feeds evenly into the comparator circuit and doesn’t affect the operation (the – input of the comparator is connected to $V_{DD}/2$ and so is R_{mbit}). Further, even if we could generate noise-free reference voltages, the comparator will exhibit an offset because the MOSFETs won’t be perfectly matched. In most sensing applications, the relative values are usually more important than the absolute values.

Example 17.4

Show that M1, M2, M3 and C_{cup} in Fig. 17.18 can be thought of as a resistor. What is the resistor’s value?

Figure 17.20 shows the circuit portion from Fig. 17.18. The current that flows in the resistor is

$$I_{avg} = \frac{V_{DD} - V_{DD}/2 - V_{THP}}{R_{SC}} \quad (17.25)$$

The amount of charge that flows during one clock cycle, if M3 is on, is

$$Q_{cup} = (V_{DD} - V_{DD}/2 - V_{THP}) \cdot C_{cup} \quad (17.26)$$

or, if this amount of charge is only allowed to flow M times out of N

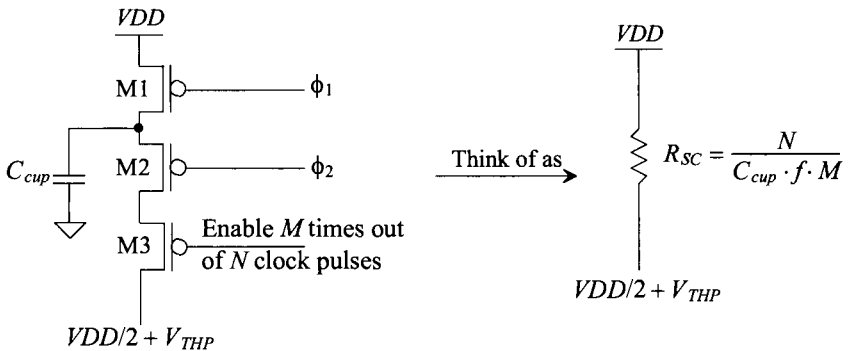


Figure 17.20 How a switched-capacitor resistor is modeled.

$$I_{avg} = \frac{Q_{cup}}{T} \cdot \frac{M}{N} = Q_{cup} \cdot f \cdot \frac{M}{N} \quad (17.27)$$

or

$$R_{SC} = \frac{N}{C_{cup} \cdot f \cdot M} \quad (17.28)$$

Noting that if M3 is always enabled (the circuit with only M1 and M2) we get a switched-capacitor resistance of

$$R_{SC} = \frac{1}{C_{cup} \cdot f} \quad (17.29)$$

which, for Fig. 17.18 and the associated discussion, is 25k. ■

Example 17.5

Show that M1, M2, and C_{cup} can be replaced with a resistor in the DSM sensing circuit of Fig. 17.18. Show the DSM's output when R_{mbit} is 50k and 200k.

The schematic for the DSM modulator is seen in Fig. 17.21. We might think that the errors due to incomplete settling would be eliminated using this scheme. However, the source of M4 will not be precisely at $V_{DD} + V_{THP}$, so an error will still be present. Again, increasing the width of M4 will lessen the error's effects.

Using the same values as used to generate Fig. 17.19b and d, we get the simulation results seen in Fig. 17.22. The big benefit of this topology over the one in Fig. 17.18 using the switched capacitor resistor is simpler design (no nonoverlapping clocks are needed) and lower power (there are not as many parasitic capacitances to charge and discharge). The current pulled from V_{DD} in Fig. 17.18 is 26 μ A when clocked at 100 MHz, while the circuit in Fig. 17.21 uses 20 μ A. The big drawback of using the simple resistor is the inability to adjust the resistance by changing the clock frequency. In a practical circuit the fabrication

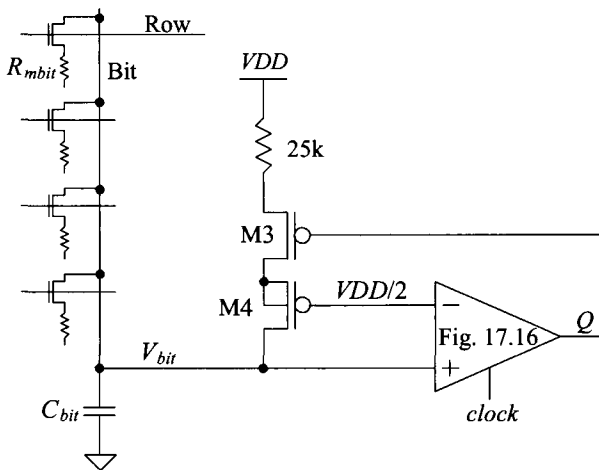


Figure 17.21 Simpler DSM for sensing resistive memory.

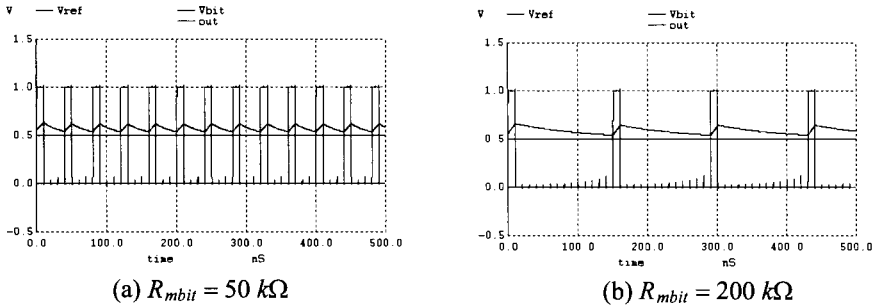


Figure 17.22 Simulation results for the circuit in Fig. 17.21.

process characteristics and temperature vary. To center the value of sense to half way between a programmed and an erased resistor value, the clock frequency can be adjusted. Adjusting the clock frequency in Fig. 17.21 simply adjusts the variation in the bit line voltage. ■

17.3 Sensing in CMOS Imagers

Another area where DSM (delta-sigma modulation) can be used for sensing is in CMOS imaging chips that acquire images in cameras or video recording. A schematic of a CMOS active pixel sensor (APS) is seen in Fig. 17.23. Light is applied through a filter (so that only red, green, or blue light passes) to the photodiode. The photodiode changes the light intensity into a charge. The charge is converted into a voltage and passed to the column line. The brighter the light, the bigger voltage change we get on the column line.

Resetting the Pixel

Prior to acquiring an image each pixel in the imaging chip is reset. This is accomplished by driving the reset row line (*ResetN*) to a voltage, $VDDP$, greater than $VDD + V_{THN}$ (with body effect). This turns M1 on and sets the voltage across the photodiode to VDD . This condition is called the *dark or reference level* of the pixel. We can then turn M3 on by driving *RowN* to $VDDP$ and, with M2 behaving as a source follower, driving the column line to the reference voltage level, V_R . This is important because each pixel in an imaging array will have slightly different characteristics. The differences in the pixels can result in speckles in the image. We can subtract this reference level from the actual measured signal level to get an accurate idea of the light intensity applied to the pixel (to eliminate the pixel gain differences).

The point here is that our DSM circuit will have to sense, and store, a reference level, V_R , at the beginning of the sense.

The Intensity Level

After we have stored the reference level voltage V_R (on a capacitor), the *ResetN* signal goes low. This allows the light striking the reverse-biased photodiode (through the color filter) to generate electron-hole pairs that cause the voltage across the diode to decrease. After a time, the information stored on the gate of M2 (the decrease in the voltage across the photodiode) is sampled on the column line (assuming M3 is on). The time between

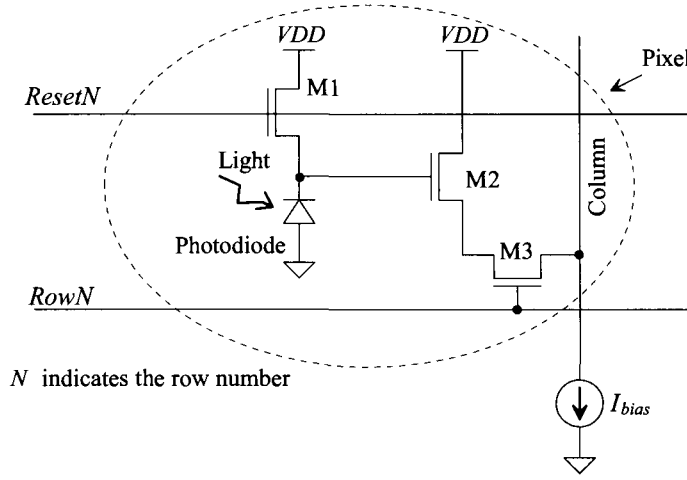


Figure 17.23 A CMOS active pixel sensor (APS).

the *ResetN* going low and the data (the column voltage, V_I , corresponding to the intensity of the light striking the photodiode) being sampled on the column line is called the *aperture time*. Note that a dark signal corresponds to a large voltage on the column line ($VDD - V_{THN}$), while a bright signal corresponds to a lower voltage (less than $VDD - V_{THN}$).

Sampling the Reference and Intensity Signals

When *ResetN* and *RowN* are high, the pixel's reference (or dark) voltage, V_R , is placed on the column line, Fig. 17.24. At this time the sample and hold reference signal, *SHR*, goes high and V_R is sampled onto a hold capacitor. Next, *ResetN* goes low and the photodiode changes light into charge. After the aperture time, the information from the photodiode (the intensity of the light) is on the column line, V_I . This voltage is then sampled and held on a hold capacitor when the signal *SHI* (sample and hold the intensity of the light) goes

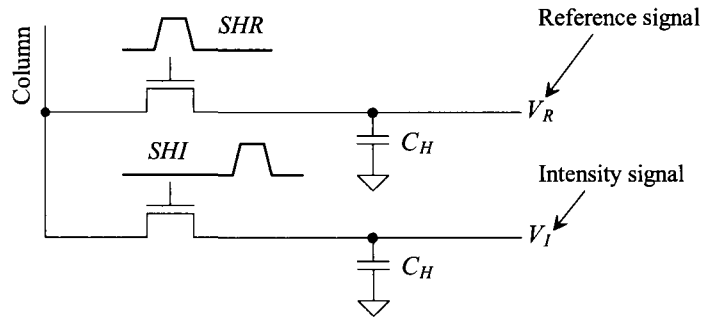


Figure 17.24 Sampling the reference and intensity signals.